Explainable Machine Learning

SHAP and Shapley Values

Reading a model's prediction as feature-by-feature contributions

Shapley Value · SHAP (SHapley Additive exPlanations)

From the fair-division rule of game theory to nine SHAP plots of a credit-scoring model.

SHAP grew from a Nobel laureate's fairness rule

Lloyd Shapley (1923–2016), American mathematician and game theorist

1953

Proposed the Shapley value, a rule for splitting a jointly earned payoff

2012

Nobel Prize in Economics with Alvin Roth, for stable allocations and market design

2017

Lundberg and Lee turned it into SHAP, a tool for explaining ML models

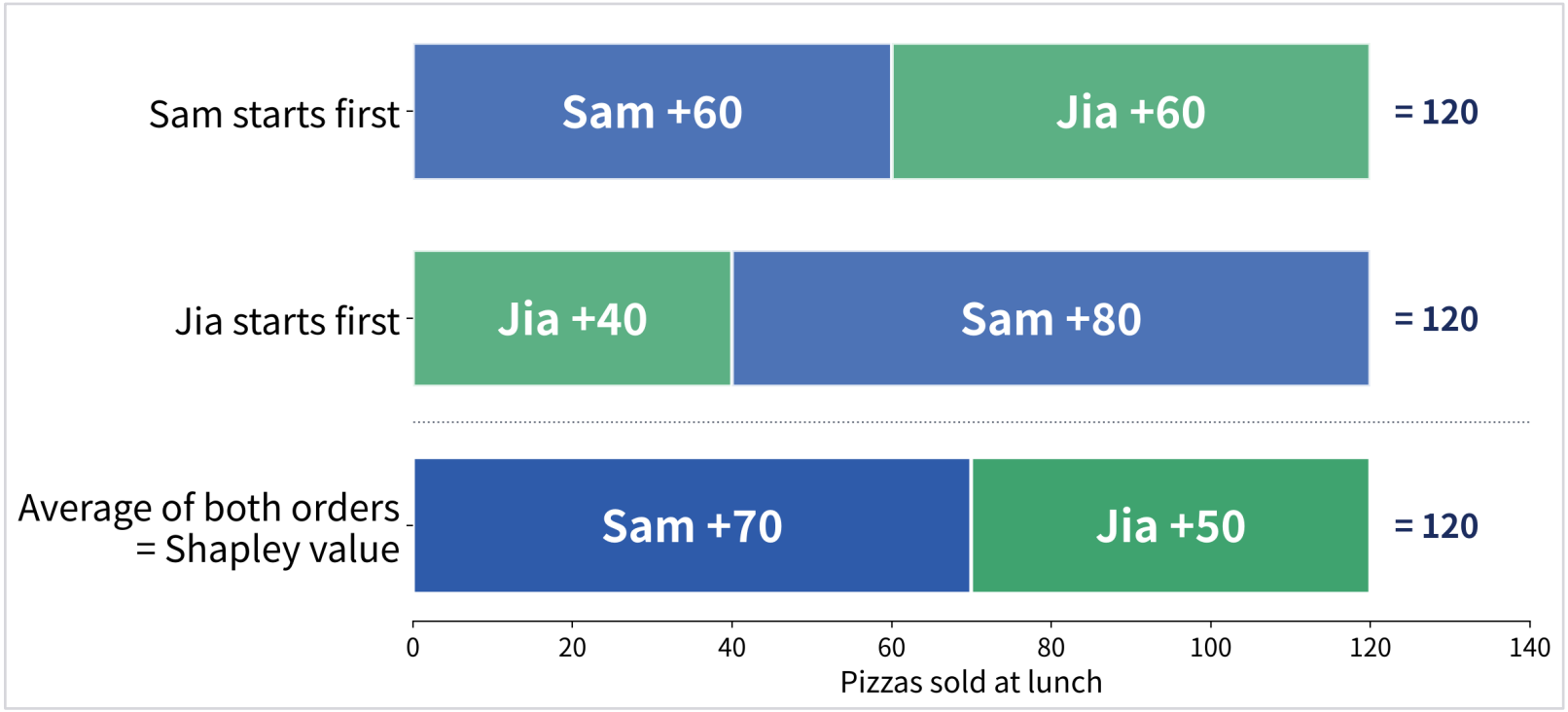
Key question: how much of a joint result should each contributor fairly receive?

Each share is the average over both starting orders

A pizza shop: Sam alone sells 60, Jia alone 40, together they sell 120

How to read

- Together they sell 20 extra
- Whoever starts second gets it
- Averaging splits it in half
- Shares sum to exactly 120



Features play staff; the prediction plays sales

Applied to a credit-score model, the same rule gives each customer feature-by-feature shares

Pizza shop	Credit-score model (SHAP)
Staff (Sam, Jia)	Features (income, debt ratio, delinquencies …)
Nobody working: 0 pizzas	Average prediction with no feature info: 592.7
Together: 120 pizzas	Customer B's prediction: 520
Shares: 70 and 50	SHAP value per feature (delinquencies −57.2 …)
Shares add up to 120	SHAP values add up to 520 − 592.7 = −72.7

Start from the average prediction and add each feature's share to reach the customer's prediction

Planted truths in synthetic data test SHAP's reading

3,000 synthetic customers · target: credit score (300–900) · XGBoost regression · TreeSHAP

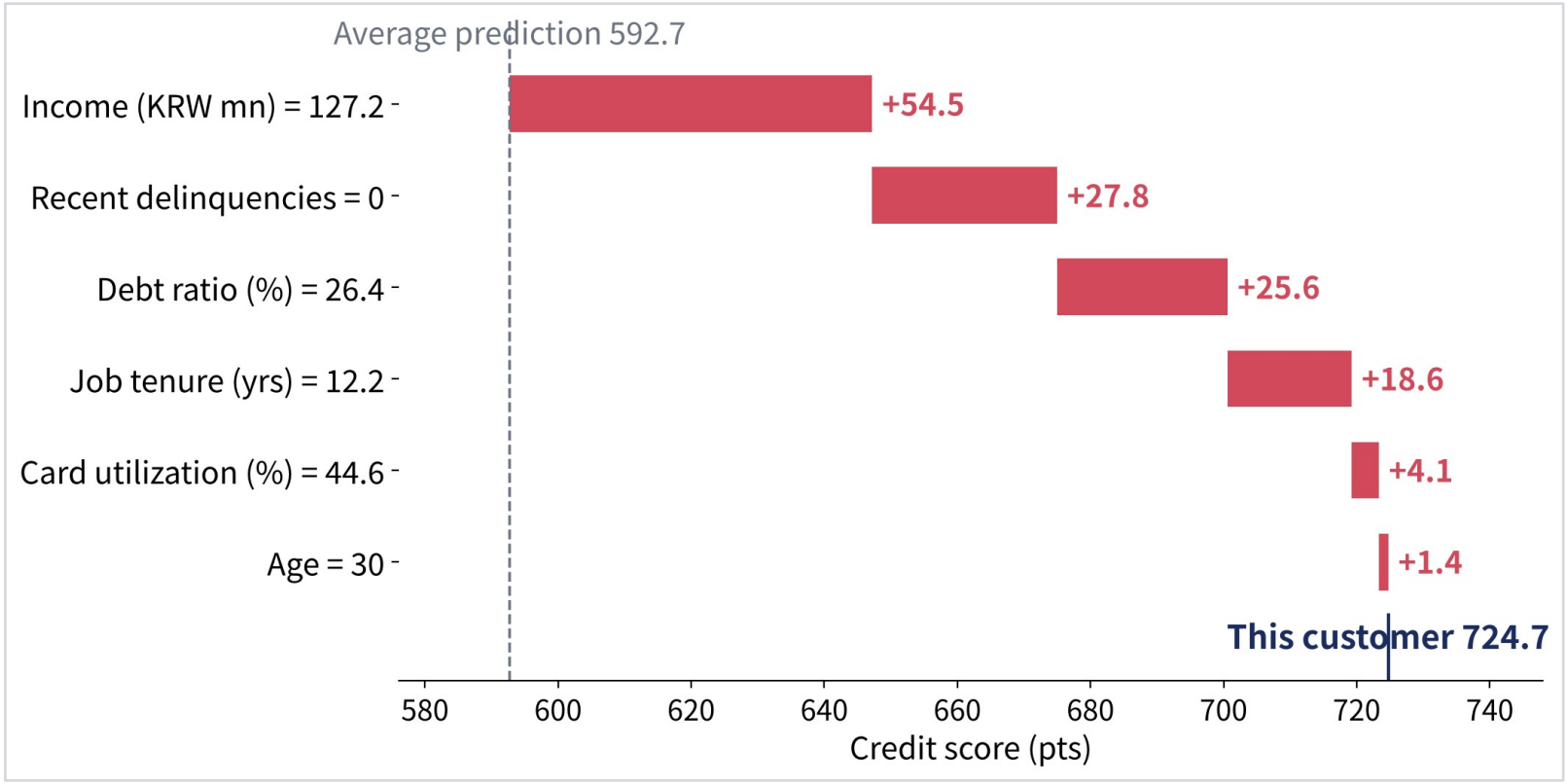
Feature	Range	Relationship planted in the data
Recent delinquencies	0–6	Large penalty each; also steepens the card-utilization penalty
Debt ratio (%)	5–140%	Penalty rises; steeper above 70%
Income (KRW mn)	15–200	Bonus rises with diminishing gains
Card utilization (%)	0–100%	Higher means penalty
Job tenure (yrs)	0–30	Bonus up to 10 years, flat after
Age	22–65	No relationship (control)

Customer A: all five features pushed the score up

Waterfall plot — the steps from the average of 592.7 to a prediction of 724.7

How to read

- Dashed line: start
- Red adds, blue subtracts
- Bar length = share
- High income: +54.5

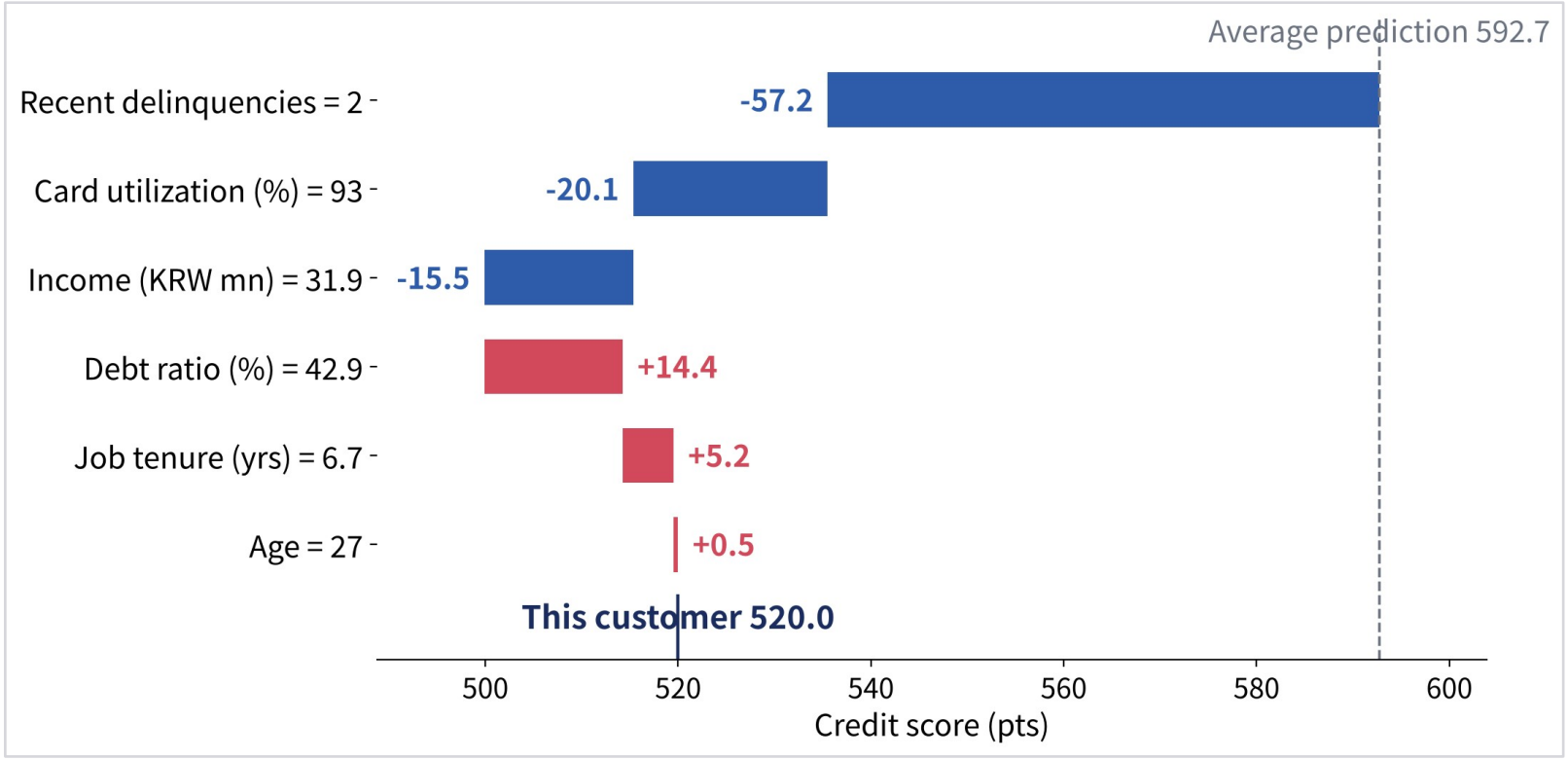


Delinquencies explain most of Customer B's low score

Prediction 520.0 = average 592.7 − 72.7

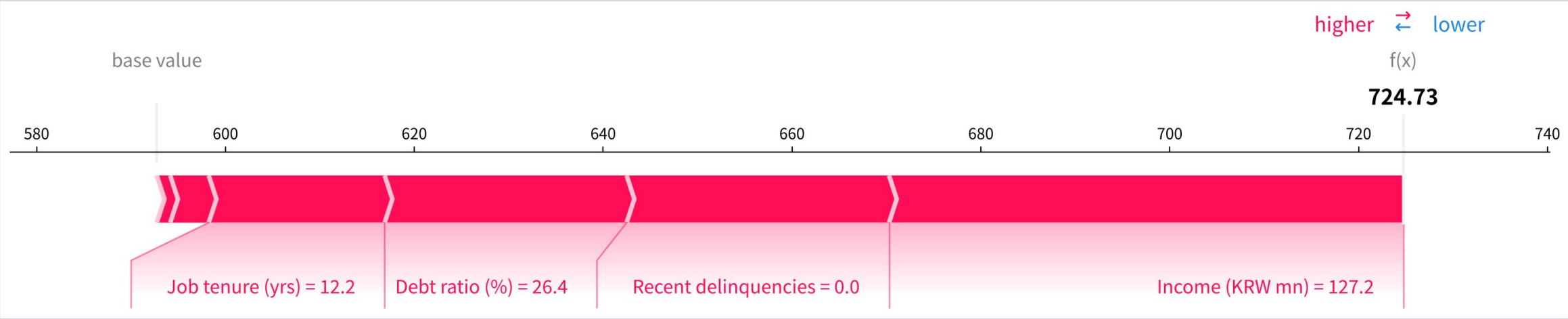
Key finding

- 2 delinquencies: −57.2
- Card use 93%: −20.1
- Debt ratio adds points
- Total exactly −72.7



A force plot is a waterfall squeezed into one line

Force plot — customer A; base value is the average prediction, f(x) is this customer's prediction



How to read

- Red arrows push the score higher
- Wider arrow, larger contribution

When to use

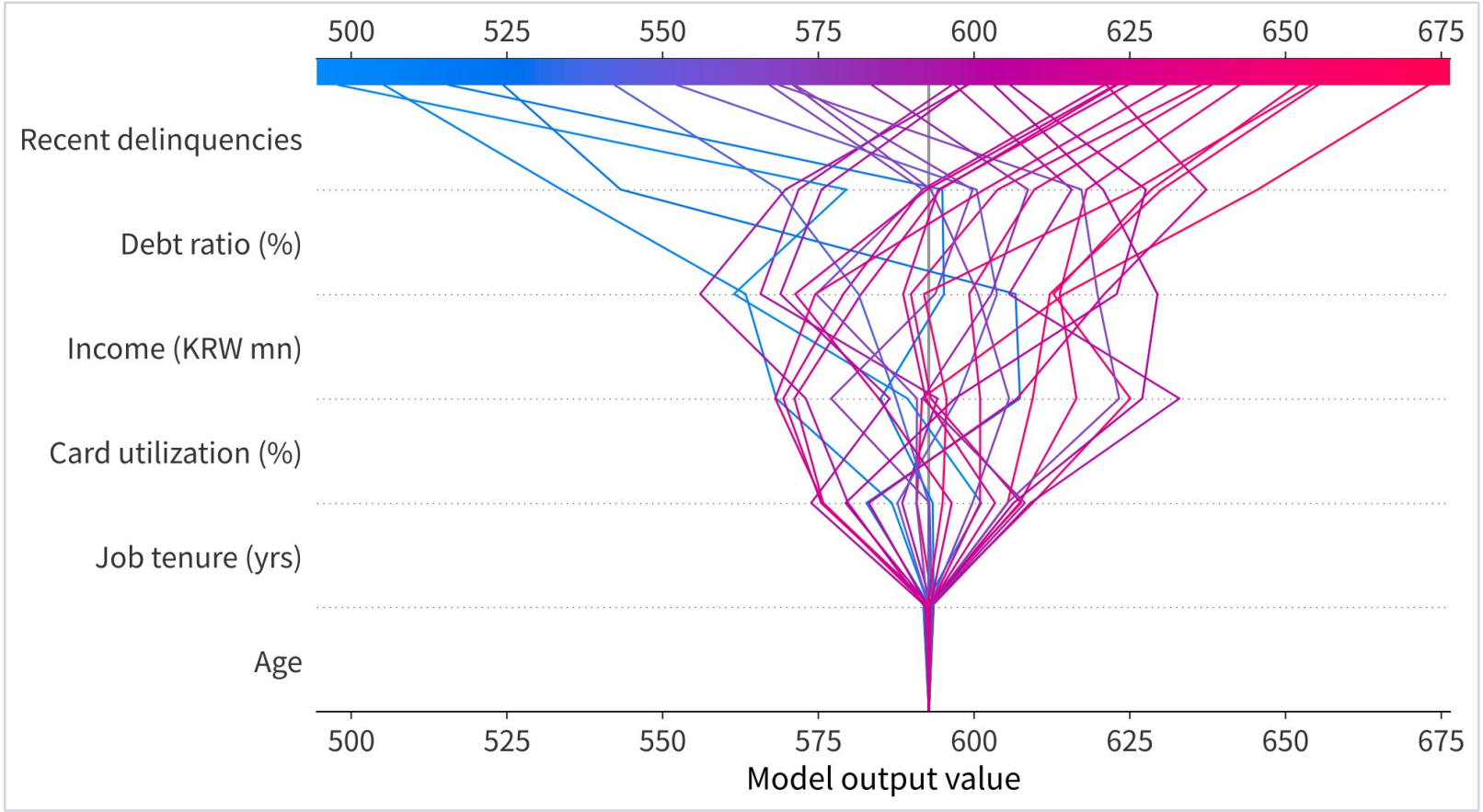
- Fits on one line of a report
- Stacked vertically, it shows many cases

A decision plot overlays many customers' paths

Decision plot — 30 customers, features added one by one from bottom to top

How to read

- Start at the average
- Contributions accumulate
- Top end = prediction
- Paths split at delinquency

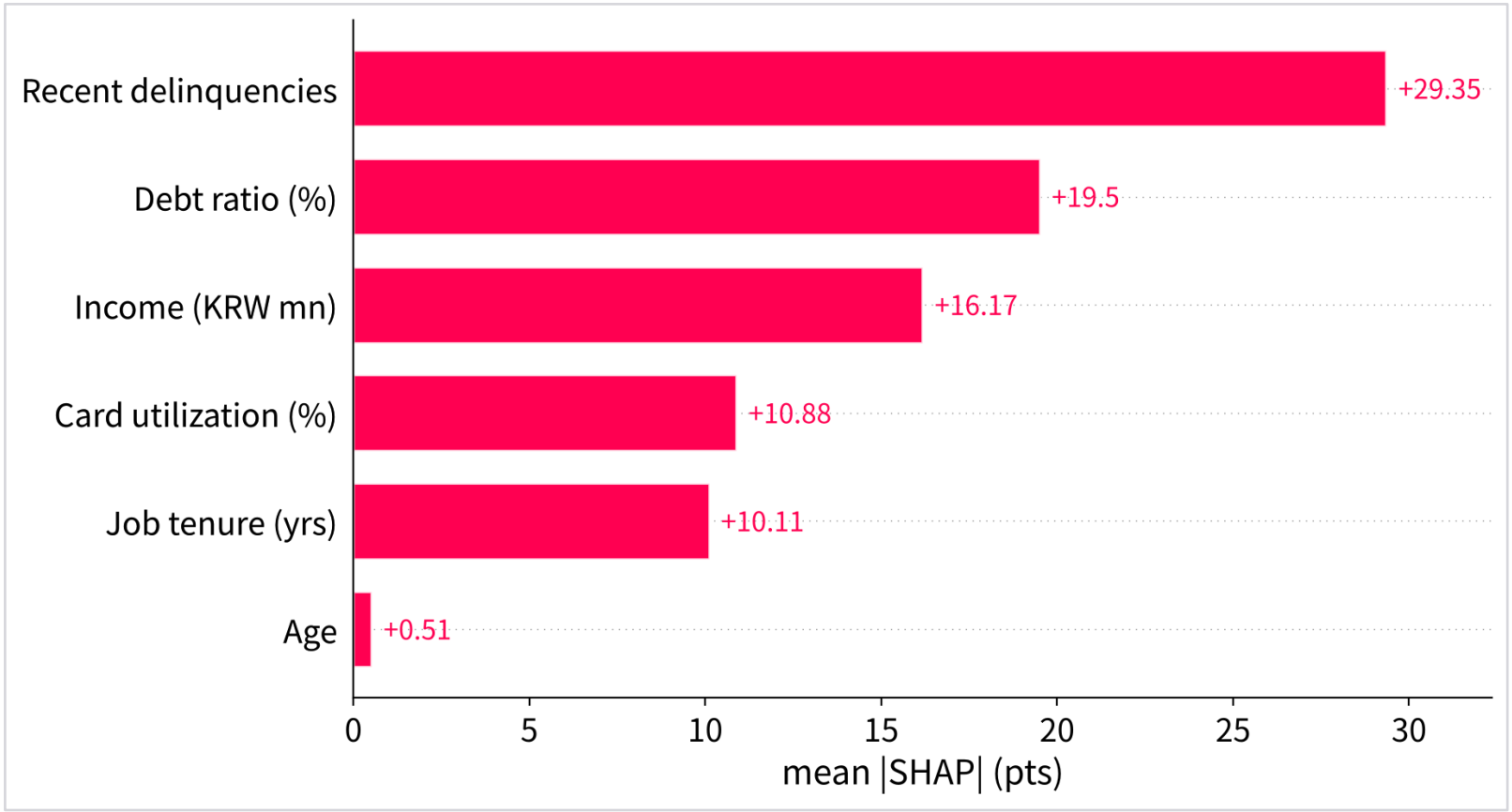


Across the model, delinquencies matter most

Bar plot — mean absolute SHAP value over 3,000 customers

How to read

- Mean |SHAP| = impact
- Direction is hidden
- Age: 0.5, irrelevant
- Matches planted order

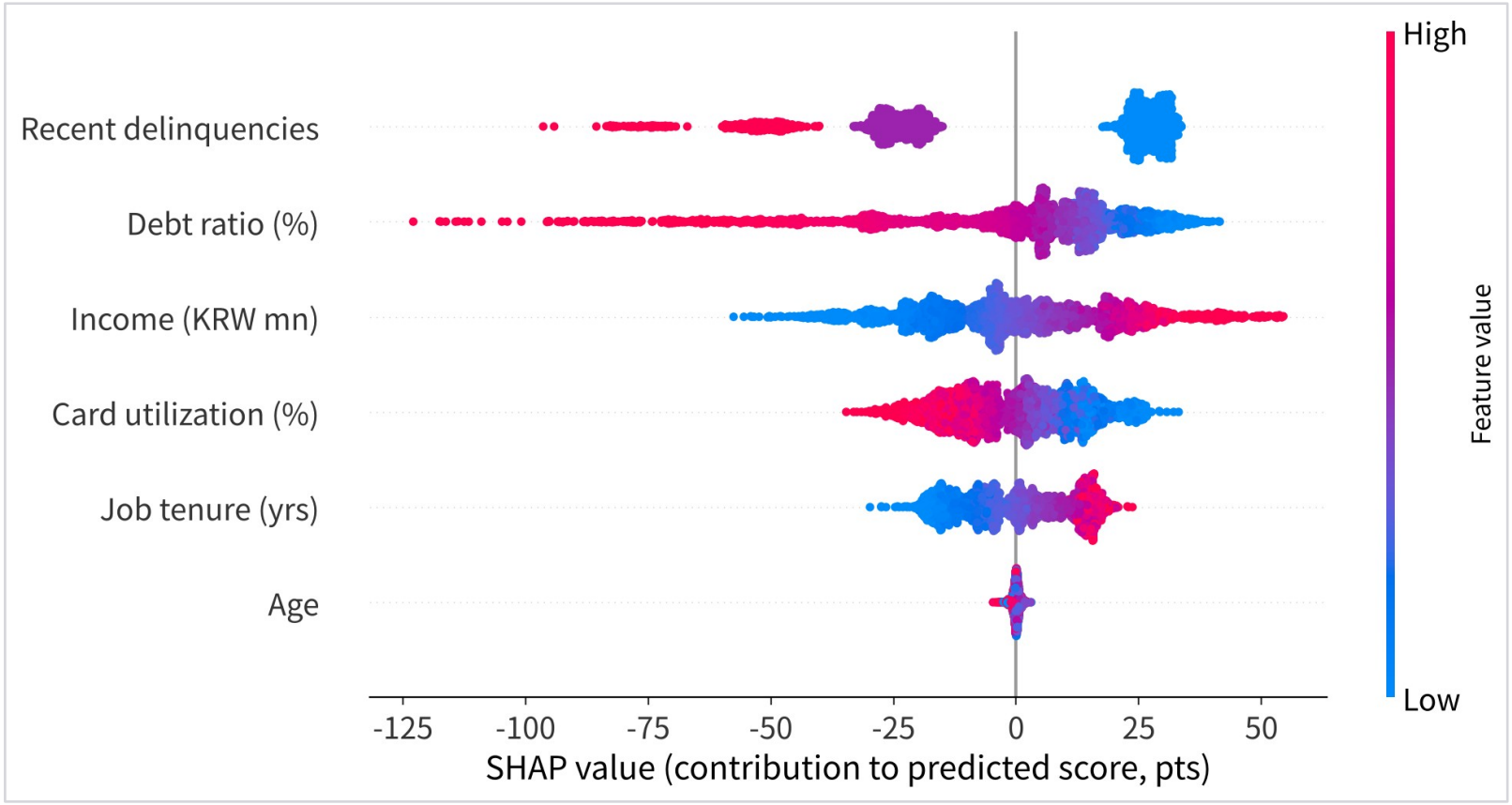


Beeswarm plots show impact size and direction

Beeswarm plot — one dot per customer, colour is the feature's value

How to read

- x-position = SHAP
- Red = high value
- Red on left = penalty
- Wide spread = impact

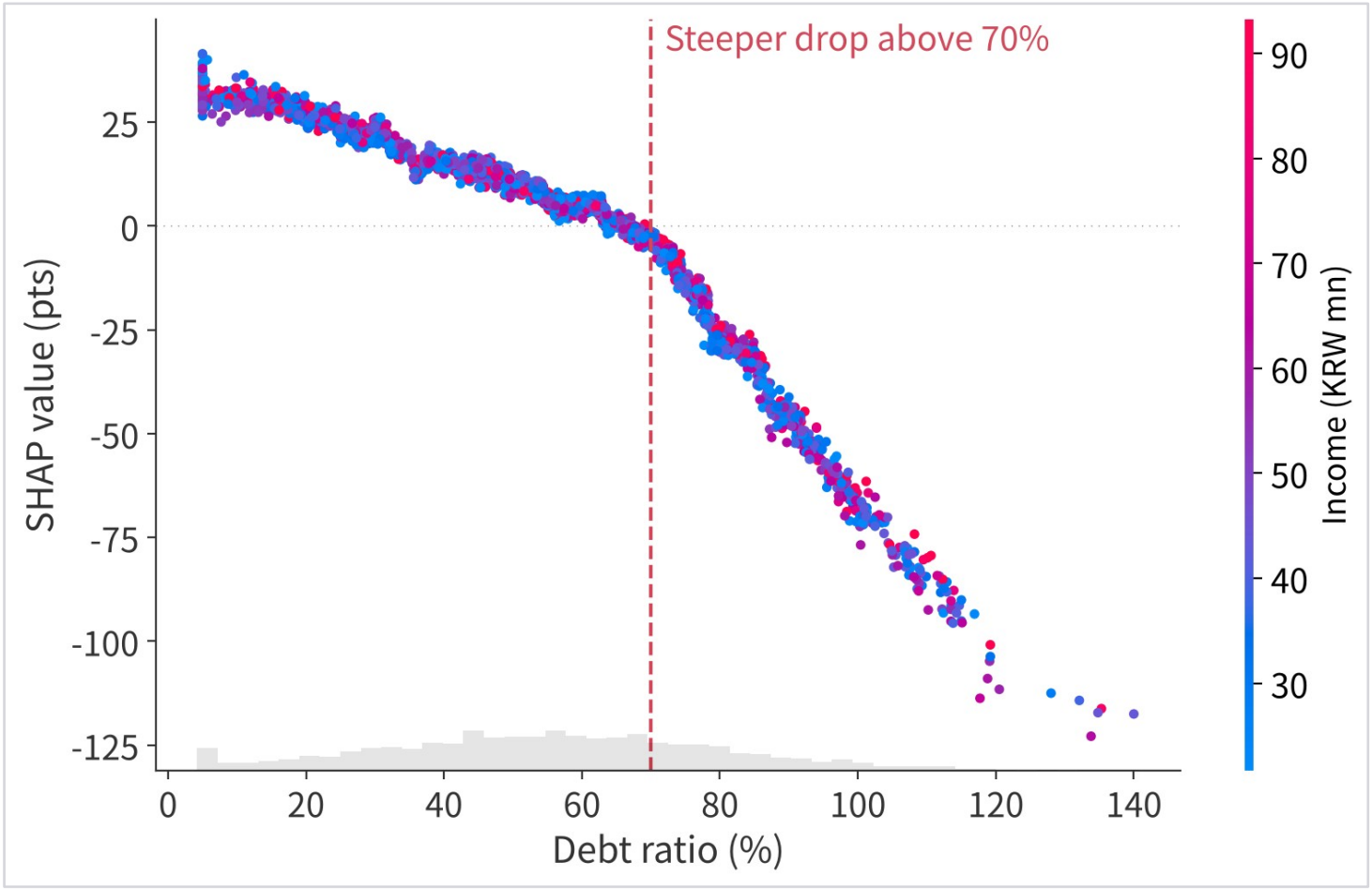


The dependence plot finds the 70% debt-ratio kink

Dependence scatter — x is debt ratio, y is its SHAP value

Key finding

- Mild penalty below 70%
- Steep penalty above
- Linear models miss this
- Grey bars: distribution

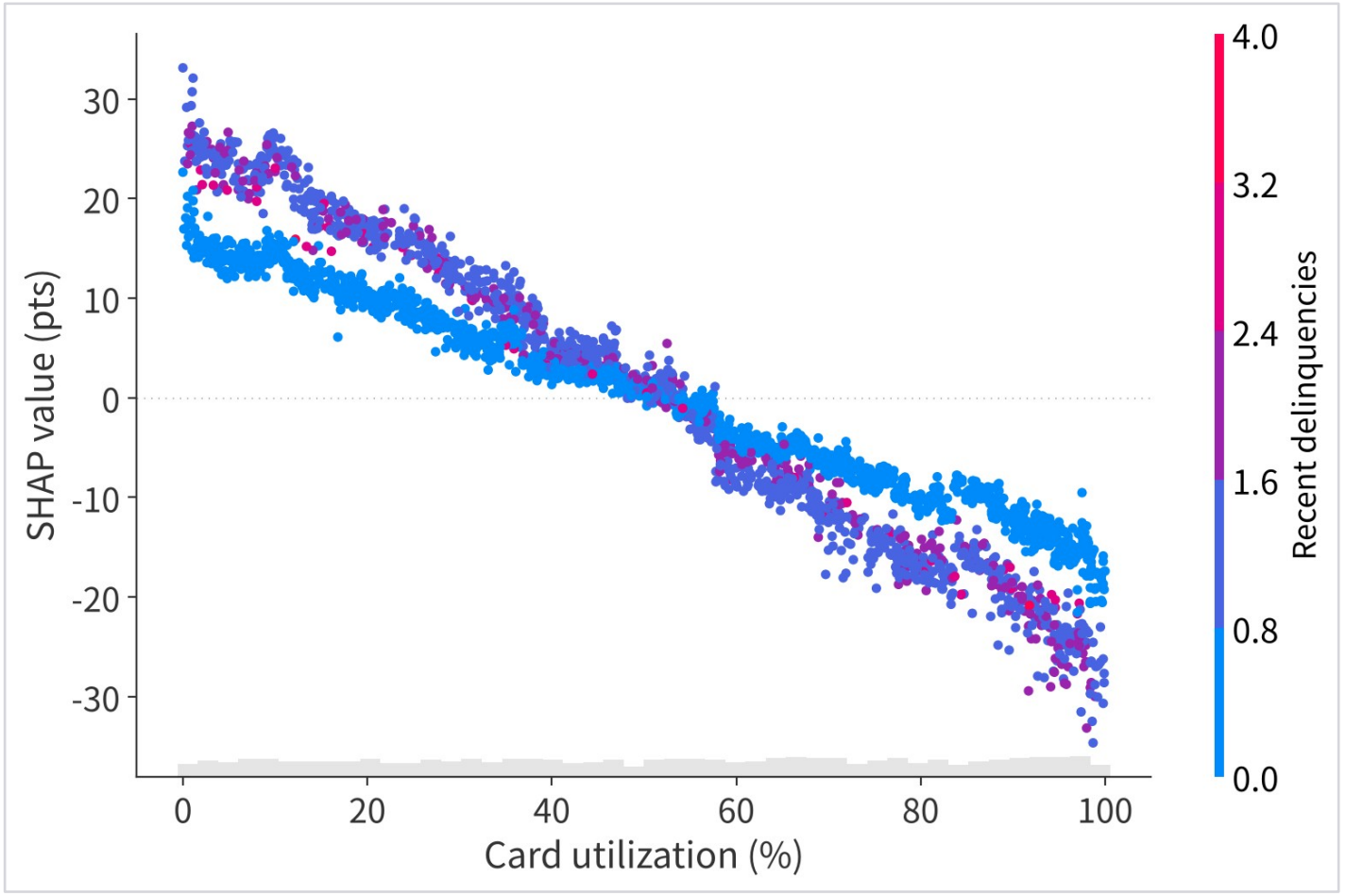


Delinquency steepens the card-utilization penalty

Same scatter coloured by delinquencies — split colours signal an interaction

Key finding

- Blue (0) is flatter
- Purple/red is steeper
- Two separate bands
- Planted effect found

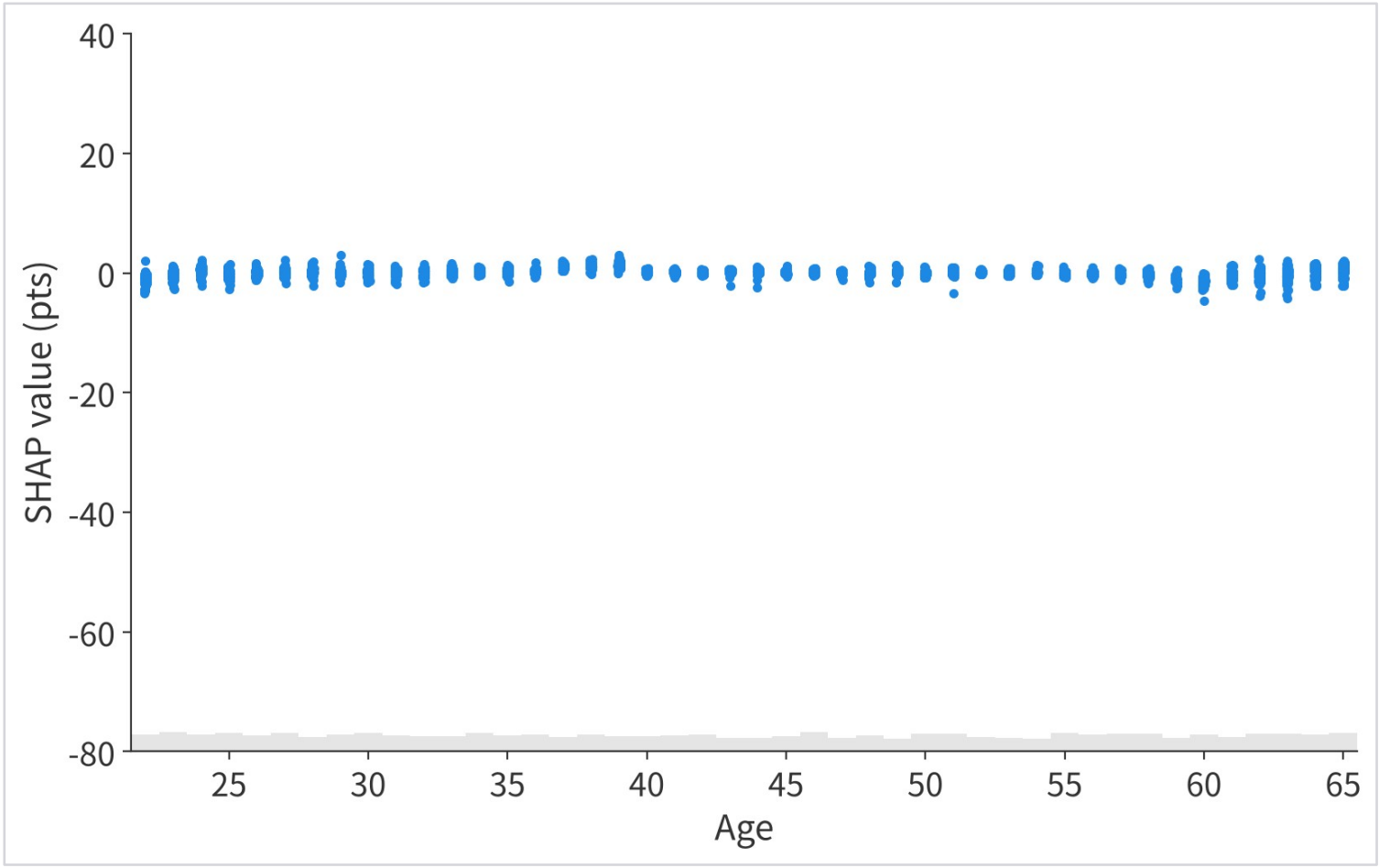


Age, with no planted effect, stays near zero

y-axis range set close to the other plots for comparison

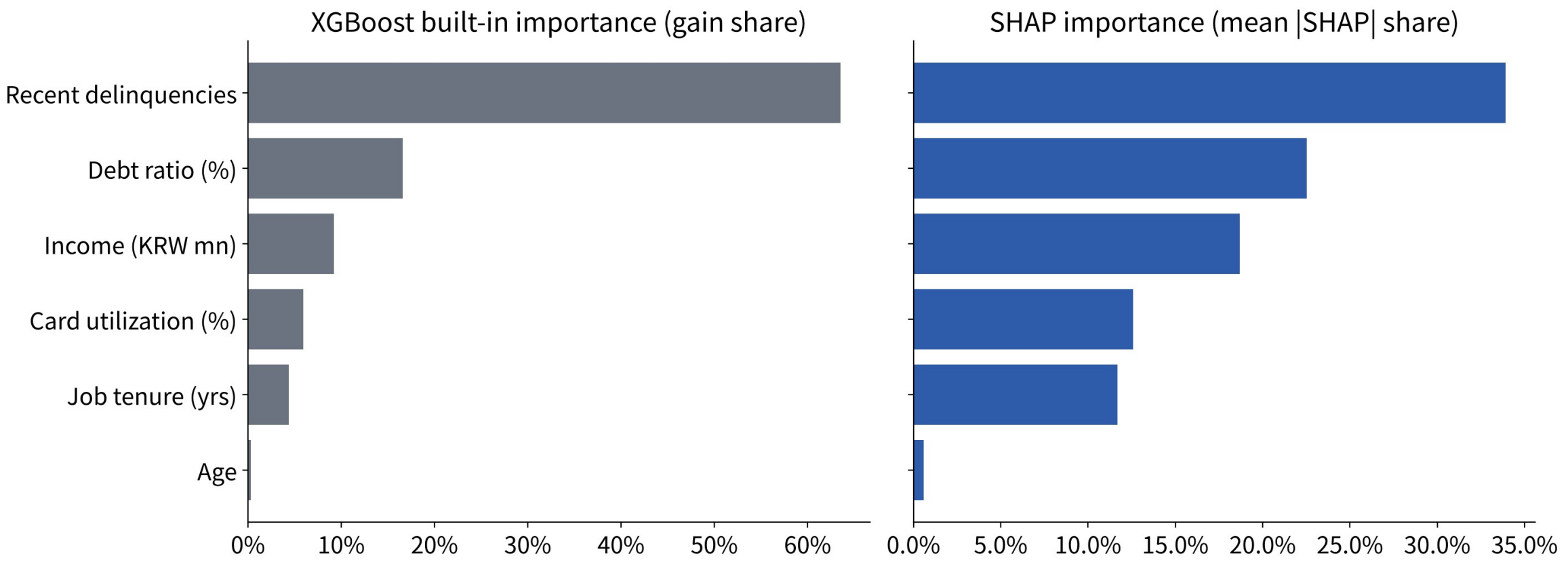
Key finding

- Noise within ±2
- No effect, no share
- A useful baseline check



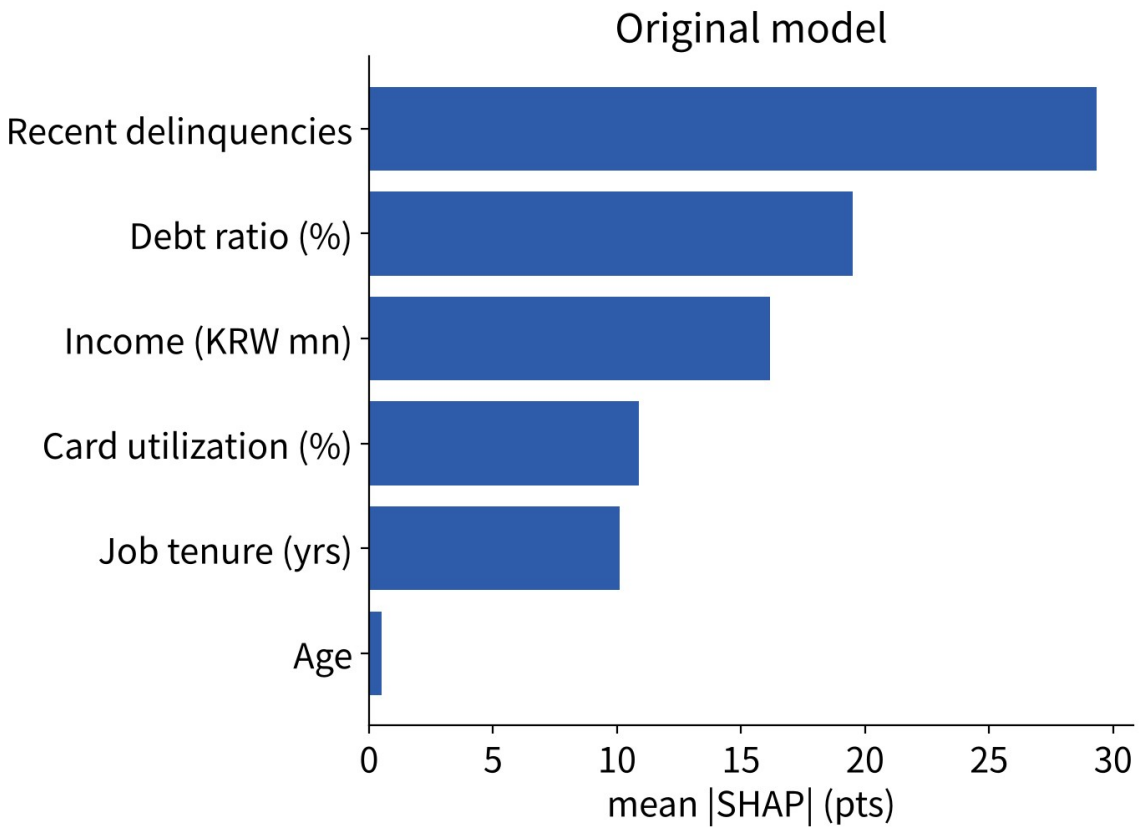
Tree gain importance over-weights delinquencies

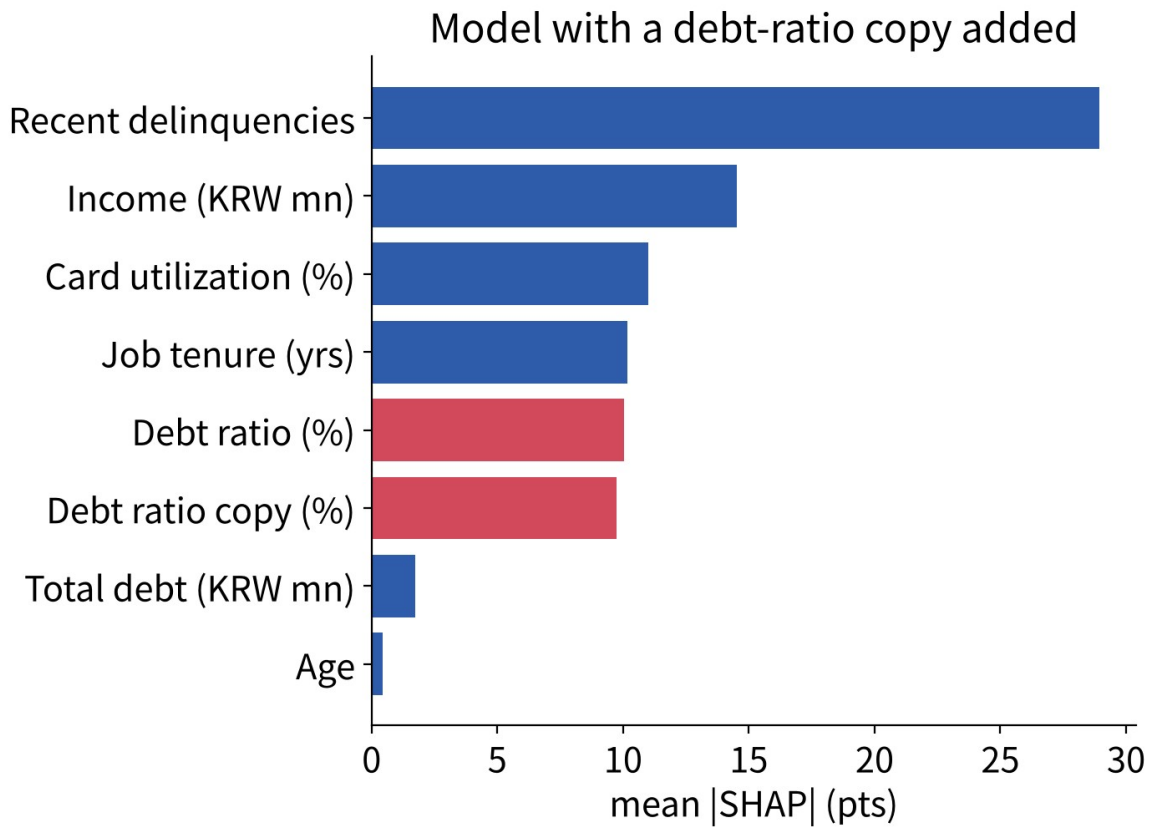
XGBoost gain gives delinquencies 63%; SHAP gives 34% and shows the other features' shares



Near-identical features split SHAP credit in half

Retrained with a noisy copy of debt ratio — its 19.5 points split into 10.0 and 9.8





SHAP shows model habits, not causation

Three rules for interpreting results

Not causal

Only how the model computes

Lower card utilization is not proof of better repayment.
SHAP describes how the model uses a feature.

"Reason", not "cause"

Correlated features share

Importance gets diluted

Several similar features each look small.
Group them or keep only one.

19.5 → 10.0 + 9.8

Baseline = background

Compared with whom?

Changing the background data changes the average and every SHAP value.
State the reference group.

All applicants or approved only?

Customer B's waterfall becomes a rejection notice

Credit practice: pick the top SHAP penalties and turn them into plain sentences

Rank	SHAP penalty	Customer notice (example)
1	2 recent delinquencies (−57.2)	Recent late payments lowered your score.
2	Card utilization 93% (−20.1)	Your card balances are high relative to limits.
3	Income 31.9 mn KRW (−15.5)	Your income is below the assessment average.

Practical point

- Positive factors (debt ratio +14.4) are never listed; reasons come only from the top penalties.

SHAP turns predictions into readable shares

1

Shapley value = average contribution over all joining orders

Shares add up to the total; a member who adds nothing gets zero.

2

Read one case with a waterfall, the model with beeswarm and bar plots

Dependence scatters reveal shapes such as kinks and interactions.

3

SHAP is not causation

Correlated features and the choice of background data change the results.